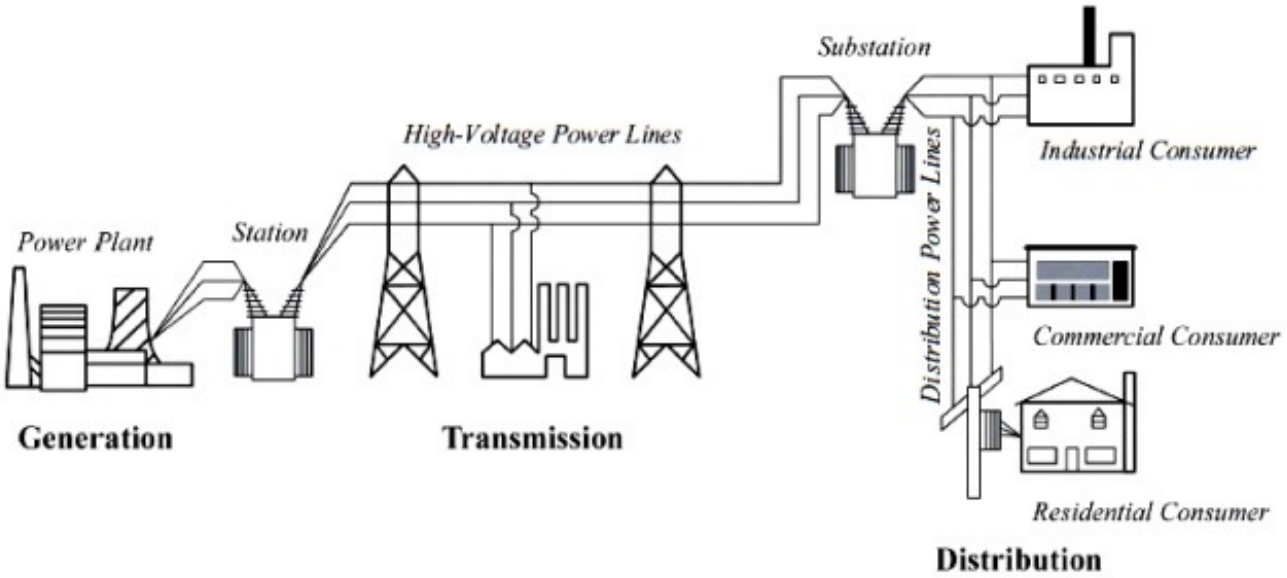
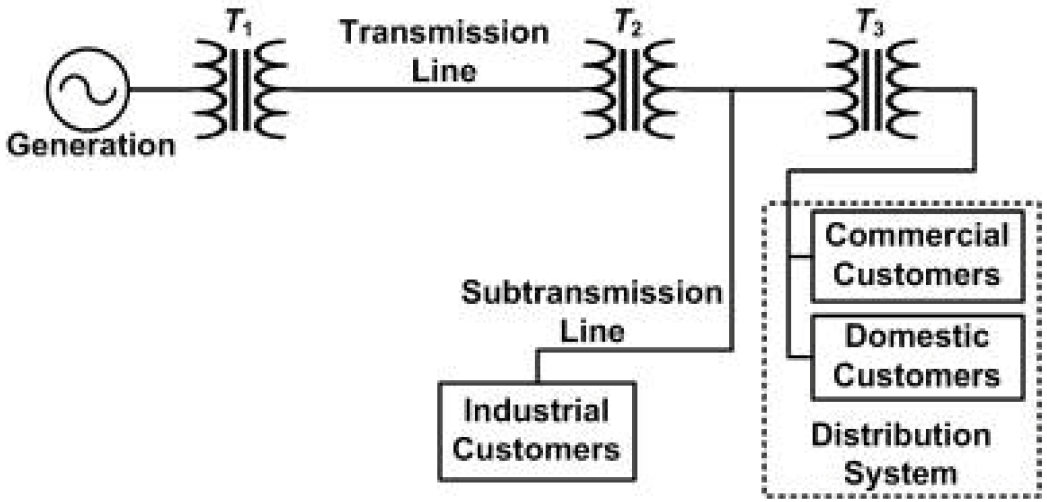
AENL 226
Power Electronics and Power Systems

المعهد الهندي للتكنولوجيا دلهي
أبوظبي
INDIAN INSTITUTE OF TECHNOLOGY DELHI
ABU DHABI

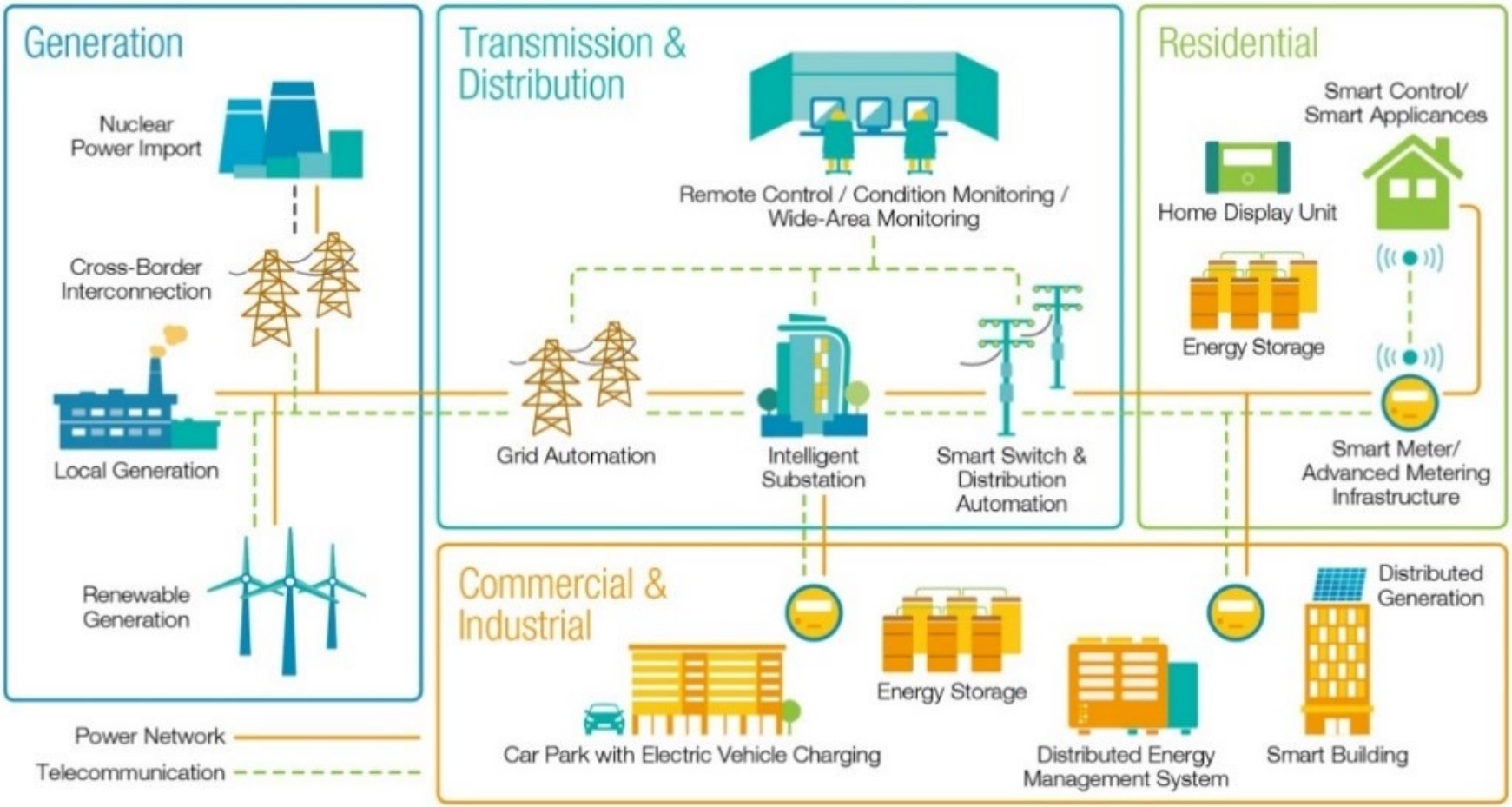


Electrical Energy Systems:
Relevance of Electrical Machines





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Relevance of Electrical Machines



https://www.clp.com.hk/en/about-clp/power-transmission-distribution/smart-grid

Electrical Energy Systems:
Relevance of Power Electronics



Integration of Renewable Energy

High Voltage DC Transmission

Flexible AC Transmission

STATCOM

SVC

TCSC

Course Learning Outcomes

- Fundamentals of semiconductor devices, power electronics converters

- Control modulation and applications of power electronic devices

- Power system modelling and operation

- Power electronic interfaces to power systems

Broad Course Outline

Lecture:

Module no.	Topic	No. of hours
1	Power electronics in energy sector, realization of generic DC-DC buck converter and analysis, concept of duty cycle and voltage averaging	4
2	Buck converter – selection of semiconductor devices, selection of inductor and capacitor; Realization and analyses of boost converter	4
3	Realization and analyses of buck-boost converter; CCM-DCM, Non-idealities in DC-DC converters and voltage gain; Multi-stage non-isolated DC-DC converters; Introduction and application of isolated DC-DC converters	5
4.	DC-AC power conversion – realization of voltage source inverter; three-phase and single-phase voltage and current source converters; Fourier analysis and Harmonic spectrum	5
5	AC to DC rectifiers – Single-phase and three-phase uncontrolled rectifiers; single-phase semi-controlled rectifiers; single-phase fully controlled rectifiers; Analyses of AC & DC side voltages & currents	5
6	Fundamentals of power generation, transmission and distribution systems, per unit representation	4
7	Load generation balance, modeling of generators, transmission lines, loads, other components	5



Broad Course Outline

8	Analysis of the performance of power systems through power flow analysis	5
9	Impact of power electronic interfaces in voltage, frequency management in power systems	5
	Course Total (14 times 'L')	**42**

Tutorials:

Module no.	Description	No. of hours
1	Voltage and current analysis of generic Buck converter	1
2	Selection of semiconductor devices, inductor and capacitor for buck and boost converters; Voltage and current analyses of boost converter	2
3	Analyses of buck-boost, Converter under CCM & DCM, Non-idealities & voltage gain; Analysis of multi-stage non-isolated DC-DC converters	2
4	Problems on inverters, rectifier operation	3
5	Power system modelling in per unit (electrical circuit-based representations)	2
6	Power flow analysis	4
	Total	**14**

Course evaluation policy

Grading Policy:

Minor	: 25%
Major Exam	: 40%
Quiz (03: best two would be considered)	: 15%
Assignments (one Power Systems, one Power Electronics)	: 20%

Attendance:

75% attendance is compulsory. Institute policy as per IITD-AD would be applicable.

Course evaluation policy

Reference Books:

- Grainger J. J. G. and Stevenson W. D., Elements of Power System Analysis, McGraw Hill Education, 2017.

- Issa Batarseh, Ahmad Harb, Power Electronics Circuit Analysis and Design, Springer, 2018

Course TA:

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